

CS4001- Professional Practices in IT

Software Contracts

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Software Contracts and Liability

What is a contract?

A contract is simply an agreement between two or more persons (the *parties* to the contract) that can be enforced in a court of law. The parties involved may be legal persons or natural persons.

Essentials for a contract

The most important and essential concerns of a contract are that:

- ▶ All the parties must intend to make a contract;
- ▶ All the parties must be competent to make a contract, they must understand what they are doing;
- ▶ There must be a 'consideration', that is, each party must be receiving something and providing something.

BESPOKE SYSTEMS

- ▶ The first type of contract, we consider are *taylor-made* or *bespoke* systems, that are custom-designed specifically designed and developed to meet the unique needs of a particular organization or individual.
- ▶ A bespoke system may consist of a single PC equipped with a word processor, a spreadsheet, and a set of macros adapted to the customer's needs or it may consist of several thousand PCs spread across 50 offices in different parts of the world, connected by a wide-area network, with large database servers and a million lines of specially written software.
- ▶ Bespoke software solution can be created for a wide range of purposes, from automating business processes to managing customer data to facilitating collaboration among employees. Because it is custom-built to fit the specific needs of a particular organization or individual, bespoke software can often provide significant advantages in terms of efficiency, effectiveness, and overall productivity.

FIXED PRICE CONTRACTS FOR BESPOKE SYSTEMS

The contract for the supply of a requirements system consists of three parts:

- ▶ A **short agreement**, which is signed by the parties to the contract: This states who the parties are and, anything that may have been said or written before does not form part of the contract.
- ▶ The **standard terms and conditions**, which are normally those under which the supplier does business.
- ▶ A set of **schedules or annexes**, which specify the requirements of this contract, including what is to be supplied, when it is to be supplied, what payments are to be made and when, and so on.

CAMBRIAN CONSULTANTS LIMITED

AGREEMENT

This agreement is made between Cambrian Consultants Limited ('the Company'), whose registered office is at
....., and
..... ('the Client'), whose registered office is at

Unless otherwise agreed by the Company in writing, this agreement, together with the terms and conditions attached as schedule A and schedule B, constitute the entire agreement between the Company and the Client and shall apply in place of, and prevail over,

- a) any other terms or conditions contained, or referred to, in correspondence or other documents, or
- b) any terms or conditions implied by trade custom and practice, or
- c) any other purported provisions or oral representations.

This agreement is made on the day of, 200 ..

Signed, for and on behalf of

Cambrian Consultants Limited,

Signed, for and on behalf of,

.....,

Name:
Director

Name:
Position:

Issues in contract

- ▶ What is to be produced
- ▶ What is to be delivered
- ▶ Ownership of rights
- ▶ Confidentiality
- ▶ Payment terms
- ▶ Penalty clauses
- ▶ Obligations of the client
- ▶ Standards and methods of working

What is to be produced

- It is clearly necessary that the contract states what is to be produced.
- requirements specification

It is important that the reference to the requirements specification identifies that document uniquely; normally this will mean quoting a date and issue number.

Problem: Any changes needed during the contract life

What is to be delivered

The following is a non-exhaustive list of possibilities:

- ▶ Source Code
- ▶ Command files/ batch files for building the executable code from the source and for installing it.
- ▶ Documentation of the design of the code.
- ▶ Reference manuals, training manuals and operations manuals.
- ▶ Software tools to help maintain the code.
- ▶ User training
- ▶ Training for the client's maintenance staff;
- ▶ Test Cases and test results.
- ▶ Deployment.

Ownership of rights

- It is important for the contract to state precisely who is to own these rights and what legal rights are being passed by the software house to the client under the contract.
- Ownership in physical items will usually pass from the software house to the client, but other intangible rights (known as intellectual property rights) making more problems.
- Software is potentially protectable by a number of intellectual property rights, such as copyright, design rights, confidentiality and trademarks.

Confidentiality

- ▶ When a major must software system is being developed, Client have to pass confidential information about its business operations to the software house. On the other hand, the software house may not want the client to divulge to other details of the program content.
- ▶ Non of the parties would like the other to disclose its secrets.
- ▶ It is usual in these circumstances for each party to promise to maintain the confidentiality of the other's secrets, and for express terms to that effect to be included in the contract.

Payment terms

A pattern of payments such as the following must be included:

- ▶ an initial payment of 15% of the contract value becomes due on signature of the contract. (Requirement gathering)
- ▶ Further stage payments become due at various points during the development, bringing the total up to 65%. (Development and testing)
- ▶ A further 25% becomes due on acceptance of the software. (Deployment)
- ▶ Final 10% becomes due at the end of the warranty period. (Beta Passed)

Calculating payments for delays and changes

- ▶ Both sides of the parties may create less or high loss regarding unable of meeting the timeline. Thus, extra work must be indulged producing extra payments.
- ▶ The contract must specify the process by which these extra payments are to be calculated.
- ▶ The contract should make provision for payments to compensate for the wasted effort, incurred, for example, when the client fails to provide information on a due date or when changes are requested that result in extra work.
- ▶ Delay payments and payments for variations to the original requirements are the most common cause of contractual disputes.

Penalty clauses

The previous subsection dealt with compensation for delays caused by the client; delays caused by the supplier are handled differently.

Delays in delivering working software are notoriously common; it might therefore be expected that contracts for the supply of software would normally include such a penalty clause. There are three reasons for this:

- ▶ Suppliers are very reluctant to accept penalty clauses and anything stronger than the example quoted above is likely to lead to reputable suppliers refusing to bid.
- ▶ If the contract is to include penalty clauses, the bid price is likely to be increased by at least half the maximum value of the penalty.
- ▶ If the software is seriously late and penalties approach their maximum, there is little incentive for the supplier to complete the work since they will already have received in stage payments as much as they are going to get.

Obligations of the client

- ▶ Client usually has some responsibilities to help get the work done smoothly. These might include:
 1. Sharing relevant documents about their business or the system's environment.
 2. Giving access to the right staff members.
 3. Providing computers and equipment for development and testing.
 4. Offering workspace, phone, and administrative support if work is done at their location.
 5. Setting up data communication as needed.
- ▶ These obligations and deadlines are usually listed in the contract. If the client doesn't meet these, they might have to pay for any delays.

Standards and methods of working

- The supplier is likely to have company standards, methods of working, quality assurance procedures, and so on, and will normally prefer to use these.
- More sophisticated clients will have their own procedures and may require that these be adhered to. In some cases, the supplier may be required to allow the client to apply quality control procedures to the project. The contract must specify which is to apply.

Project meeting

- Regular progress meetings are essential to the successful completion of a fixed price contract, and it is advisable that standard terms and conditions require them to be held.
- The minutes of progress meetings, duly approved and signed, should have contractual significance in that they constitute evidence that milestones have been reached (so that stage payments become due) and that delay payments have been agreed.

Project managers

- Each party needs to know who, of the other party's staff, has day-to-day responsibility for the work and what the limits of that person's authority are.
- The standard terms and conditions should therefore require each party to nominate, in writing, a project manager.
- The project managers must have at least the authority necessary to fulfil the obligations that the contract places on them.
- It is particularly important that the limits of their financial authority are explicitly stated, that is, the extent to which they can authorize changes to the cost of the contract.

Acceptance procedures

- Acceptance procedures are the criteria by which successful completion of the contract is judged.
- The essence of the acceptance procedure is that the client should provide a fixed set of acceptance tests and expected results and that successful performance of these tests shall constitute acceptance of the system.
- The tests must be provided at or before the start of the acceptance procedure; within reason, there may be as many tests as the client wishes, but extra tests cannot be added once the test set has been handed over. (Requirement gathering)
- The purpose of this restriction is to ensure that the acceptance procedure can be completed in reasonable time.
- The term should be included in the contract that who shall be present when the tests are carried out and what happens if the tests are not completed successfully. (Exceptional Handling)

Warranty and maintenance

- Once the product has been accepted (Deployment done), beta testing required, typically 90 days. Any errors found in the software and reported within this period will be corrected free of charge. This clause is, subject to negotiation, reducing or eliminating the warranty period will reduce the overall cost of the contract and prolonging the period will increase it.
- Once the warranty period is over, the supplier may offer, or the client demand, that maintenance will continue to be available on request (Support). Since such maintenance is likely to involve enhancement of the software rather than simply correction of faults, the resources required are unpredictable –the client almost certainly does not know what enhancements will be required in two years' time.
- For this reason, a fixed price for the maintenance will not be appropriate. Maintenance will therefore usually be charged on a time and materials basis

Inflation

In long-term projects or project commitments for maintenance, the supplier will wish to ensure protection against the effects of unpredictable inflation. For that inflation clause added which allows charges to be increased in accordance with the rise in costs.

The clause should state how often (once a year, twice a year) charges can be increased and how the effect on the overall price is to be calculated.

Indemnity

- It could happen that, because of the client's instructions, the supplier is led unwittingly to infringe the intellectual property rights of a third party or that, through carelessness or dishonesty, the supplier provides a system which infringes such rights - perhaps through using proprietary software as a component of the system delivered.
- For this reason, it is advisable to include a clause under which each party indemnifies the other (that is, guarantees to cover any costs the other party becomes subject to) for liability arising from its own faults in this respect.

Termination of contract

- There are many reasons to terminate a contract before it has been completed. For example:
 - Client to be taken over by another company that already has a system of the type being developed,
 - For a change in policy on the part of the client to mean that the system is no longer relevant to its needs.
- It is essential to make provision in contract for terminating the work in an amicable manner. This usually means that the supplier is to be paid for all the work carried out up to the point where the contract is terminated, together with some compensation for the time needed to redeploy staff on other revenue earning work. The question of ownership of the work so far carried out must also be addressed.

Software Arbitration

Contracts often contain a clause that in the event of a dispute that they cannot solve themselves, the parties agree to accept the decision of an independent arbitrator.

In the case of computer-related contracts, arbitrator is to be appointed either by the President of the BCS or by the President of the IEE. Both bodies maintain lists of qualified arbitrators who have the necessary technical understanding.

Arbitration

- If arbitration is required, it will take place in accordance with the Arbitration Act 1996.
- This Act of Parliament lays down a set of rules for arbitration that cover many eventualities, and reference to it avoids the need to spell these out in detail.
- Most of the provisions of the Act are optional, in the sense that they come into effect only if the contract contains no alternative provision.

Applicable law

Where the supplier and the client have their registered offices in different legal jurisdictions or performance of the contract involves more than one jurisdiction, it is necessary to state under which laws the contract is to be interpreted.

CONSULTANCY AND CONTRACT HIRE

CONTRACT HIRE

- ▶ Contract hire is an arrangement in which the supplier agrees to supply the customer with the services of a certain number of staff at agreed daily or hourly charge rates.
- ▶ The customer takes responsibility for managing the staff concerned. Either party can terminate the arrangement at short notice.
- ▶ The supplier's responsibility is limited to providing suitably competent people and replacing them if they become unavailable or are unsuitable by the client.
- ▶ The contract for such an arrangement is usually simple. Payment is based on a fixed rate for each day worked. the rate depends on the experience and qualifications of the staff.
- ▶ Ownership of intellectual property rights generated in the course of the work may be needed to be addressed.

CONSULTANCY

- ▶ Consultants are experts who are called in by an organization to assess some aspect of its operations or its strategy and to make proposals for improvements.
- ▶ **Consultancy Project:**
A consultancy project involves providing expert advice or reports for a fixed price. These projects typically have simpler contracts compared to fixed-price contracts due to lower financial stakes and the challenge of objectively assessing the quality of reports.
- ▶ **Approaches:**
 1. **Simplified Contracts:** Used because financial risks are minimal for both parties.
 2. **Reliance on Reputation:** Ensures quality through the supplier's desire to maintain their professional reputation.

Key Aspects of a Consultancy Contract

1. Confidentiality:

1. Consultants gain access to sensitive company information.
2. Safeguards are necessary to prevent misuse for personal profit.

2. Terms of Reference:

1. Contracts must clearly define the scope of the consultancy work.
2. Disagreements may arise if consultants want to expand beyond the original scope.

3. Liability:

1. Consultants often limit liability for losses caused by their advice.
2. Clients may demand professional liability insurance to mitigate risks.

4. Control Over Final Report:

1. Contracts usually require a draft report to be reviewed by the client.
2. Clients can request changes, after which the consultant submits the final version.

Time and Materials

A time and materials contract (often referred to as a 'cost plus' contract) is somewhere between a **contract hire agreement** and a **fixed price contract**. The supplier agrees to undertake the development of the software in much the same way as in a fixed price contract, but payment is made based on the costs incurred, with labor charged in the same way as for contract hire. The supplier is not committed to completing the work for a fixed price, although a maximum payment may be fixed beyond which the project may be reviewed.

OUTSOURCING

Outsourcing, sometimes known as facilities management, is the commercial arrangement under which a company or organization (the customer) hands over the planning, management and operation of certain functions to another organization (the supplier).

IT outsourcing contracts are inherently complex and depend very much on individual circumstances. It is not appropriate to go into detail here about such contracts but the following is a list of just some of the points that need to be addressed:

KEY POINTS OF OUTSOURCING

- ▶ how is performance to be monitored and managed.
- ▶ what happens if performance is unsatisfactory.
- ▶ which assets are being transferred.
- ▶ staff transfers.
- ▶ audit rights.
- ▶ contingency planning and disaster recovery.
- ▶ intellectual property rights in software developed during the contract.
- ▶ duration of the agreement and termination provisions.

HEALTH AND SAFETY

The ones that are of particular concern to software engineers are:

- ▶ provision and maintenance of safe plant.
- ▶ provision and maintenance of safe systems of work.
- ▶ provision of such information, instruction, training, and supervision, as necessary.
- ▶ ensuring the workplace is maintained in a safe condition.
- ▶ provision and maintenance of a safe working environment and adequate welfare arrangements.

The Act also requires employers to ensure that their activities do not expose the public to risks to their health and safety. Manufacturers of equipment to be used at work also have a responsibility to ensure that it is safe.

Failure to comply with the Health and Safety at Work Act is a criminal offence and, in serious cases, can lead to criminal proceedings being taken against individuals.